

LETTER TO THE EDITOR

Reply: Visually-sensitive networks in essential tremor: evidence from structural and functional imaging

**Derek B. Archer,¹ Stephen A. Coombes,¹ Winston T. Chu,^{1,2} Jae Woo Chung,¹
Roxana G. Burciu,¹ Michael S. Okun,³ Aparna Wagle Shukla³ and
David E. Vaillancourt^{1,2,3}**

1 Laboratory for Rehabilitation Neuroscience, Department of Applied Physiology and Kinesiology, University of Florida, Gainesville, FL, USA

2 Department of Biomedical Engineering, University of Florida, Gainesville, FL, USA

3 Department of Neurology and Center for Movement Disorders and Neurorestoration, College of Medicine, University of Florida, Gainesville, FL, USA

Correspondence to: David E. Vaillancourt, PhD
University of Florida
Laboratory for Rehabilitation Neuroscience
Department of Applied Physiology and Kinesiology
PO Box 118205
Gainesville, FL 32611-8205, USA
E-mail: vcourt@ufl.edu

Sir,

We thank Tuleasca and colleagues (2018b) for their thoughtful commentary on our manuscript, in which we acutely increased visual feedback and found this exacerbated the amplitude of tremor in patients with essential tremor. Exacerbated tremor was associated with abnormal changes in the blood oxygenation level-dependent (BOLD) signal within the cerebello-thalamo-cortical pathway and extended to visual and parietal areas (Archer *et al.*, 2018). This work is consistent with prior work from our group, which suggested that BOLD amplitude in visual areas was associated with tremor amplitude (Neely *et al.*, 2015). Our findings that visual areas may play a role in the severity of tremor are complimented by several studies by Tuleasca *et al.* (Tuleasca *et al.*, 2017, 2018a, c). While collectively all of the aforementioned studies point to the idea that there is a widespread functional network related to the amplitude of tremor in essential tremor, one point of departure is whether structural changes exist in essential tremor.

Modulation in functional MRI activity in essential tremor

A prior functional MRI study from our group showed that the BOLD signal in the visual cortex (V1/V2) and primary motor cortex (M1) was related to force oscillations in the 3–8 Hz range (Neely *et al.*, 2015). These findings, in addition to several behavioural observations, suggest that visuomotor processes could play a role in the amplitude of tremor. In our recent report (Archer *et al.*, 2018), we extended these findings by showing that changes in BOLD signal in regions within the cerebello-thalamo-cortical pathway and the extrastriate visual areas (V3/V5) change with an increase in visual feedback. Moreover, changes within a network comprised of the primary motor and somatosensory cortex (M1/S1), inferior parietal lobule (IPL), extrastriate visual cortex (V3/V5), and cerebellar lobule VI are associated with the changes in the amplitude of tremor in response to increases in visual feedback. Importantly, this network was not specific to our behavioural manipulation

and was also associated with clinical tremor (i.e. Fahn-Tolosa-Marin Tremor Rating Scale).

While our current study focused on acute changes in the BOLD signal and the amplitude of tremor, the functional MRI study by Tuleasca *et al.* (2018a) examined changes in resting state functional MRI activity and clinical tremor. Specifically, they performed left unilateral stereotactic radiosurgical thalamotomy (SRS-T) on 17 individuals, all of whom were scanned with resting state functional MRI at baseline and at 1 year following surgery. They found that tremor score improvement on the treated hand was associated with interconnectivity strength in the salience network with the claustrum and putamen. Moreover, interconnectivity strength of the right visual association area with the bilateral motor cortices, frontal eye fields, and cerebellum lobule VI was associated with tremor score on the treated hand improvement. Together, our study and that of Tuleasca *et al.* show that the severity of tremor extends beyond the cerebello-thalamo-cortical pathway. Tuleasca *et al.* suggest that the corpus callosum, basal ganglia, and/or the cerebellum may be a mediator of the severity of tremor—our results support this hypothesis. Specifically, our findings support the hypothesis that the cerebellum is involved in the severity of tremor, but we found no direct structural or functional associations of the corpus callosum or basal ganglia.

Structural changes in essential tremor

One departure between work from our group (Archer *et al.*, 2018) and work from Tuleasca *et al.* (2017, 2018c) is that we find functional deficits in the absence of structural deficits, while Tuleasca *et al.* find structural deficits in essential tremor. A few points can be made on these differences. First, our studies implemented an acute visual feedback manipulation task to exacerbate the amplitude of tremor, whereas the studies by Tuleasca *et al.* had a longitudinal design where they studied the effects of radiosurgery of the ventro-intermediate nucleus (VIM). In our current study, we found no structure-function relationship using either voxel-based morphometry (VBM) or diffusion MRI probabilistic tractography analyses (Archer *et al.*, 2018). Tuleasca *et al.* mention two of their VBM studies. In the first VBM study, they found that a decrease in grey matter density at 1 year in Brodmann area 19 (BA19) and parahippocampal place area after VIM radiosurgery was related to higher improvements in tremor score on the treated hand (Tuleasca *et al.*, 2017). In their 2018 VBM study, they found that higher baseline grey matter density in BA18 was associated with better improvement in tremor score on the treated hand at 1 year (Tuleasca

et al., 2018c). The key difference between the study from our group and those by Tuleasca *et al.* is that we performed an acute behavioural manipulation whereas Tuleasca *et al.* performed a longitudinal study over 1 year. Second, cohort selection could be a major driver of mixed results, as there is substantial heterogeneity across essential tremor patients such that essential tremor can be viewed as a family of diseases (Louis, 2005). Recently, it has been suggested that essential tremor could be a placeholder for diagnoses that we do not fully understand, and that it is fine to use until there is a better explanation (Fasano *et al.*, 2018). For this reason, it is possible that differences in cohort enrolment play a role in the different findings regarding structural atrophy.

Collectively, our current study and the studies from Tuleasca *et al.* show that there is a widespread functional network that is related to the amplitude of tremor. Future studies with larger cohorts and multimodal structural imaging could further elucidate the structural deficits in essential tremor.

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